

Wollo University
Kombolcha Institute of Technology
College of Informatics
Department of Information Technology
System and Network Administration Lab Assignment (25%)

Part A: Windows Server 2016

1. Active Directory and User Management

1. Install Active Directory Domain Services (AD DS) and promote the server to a domain controller.
2. Create a new user account and assign it to an organizational unit (OU).
3. Set password policies and enforce password expiration for all users.
4. Create and apply a Group Policy Object (GPO) to restrict USB access for certain users.

2. File and Storage Management

1. Create a shared folder and set NTFS and share permissions for different users.
2. Configure disk quotas on a drive to limit user storage.
3. Enable and configure Distributed File System (DFS) for replication between two servers.

3. Network Services

1. Install and configure the DHCP server to assign IP addresses automatically.
2. Install and configure DNS, and create a forward lookup zone.
3. Configure a basic VPN server using Routing and Remote Access (RRAS).

4. Security and Backup

1. Configure Windows Firewall rules to allow only specific services.
2. Implement Windows Server Backup to schedule regular backups of critical folders.
3. Enable BitLocker on a volume and test recovery keys.

Part B: Ubuntu Server (Linux)

1. User and Group Management

1. Create users and assign them to groups with specific permissions.
2. Configure sudo privileges for administrative users.
3. Lock and unlock user accounts using command-line tools.

2. File System and Storage

1. Mount and unmount a new disk and create an ext4 filesystem.
2. Set file permissions using chmod, chown, and chgrp.
3. Configure Logical Volume Management (LVM) to expand storage on a volume.

3. Network Configuration

1. Configure a static IP address and verify network connectivity using ping and netstat.
2. Set up SSH server for remote administration and test login from a client machine.
3. Configure a DHCP client and verify that IP is obtained from the DHCP server.

4. Services and Security

1. Install and configure Apache web server and create a simple HTML page.
 2. Set up a firewall using ufw to allow only HTTP, HTTPS, and SSH traffic.
 3. Configure automatic updates using apt and schedule regular backups with rsync.
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Part C: Combined Tasks

1. Implement cross-platform authentication using LDAP between Windows Server and Ubuntu Server.
2. Configure file sharing between Windows and Ubuntu servers using Samba.
3. Monitor network traffic and server performance using tools like Wireshark, top, and perfmon.